



Inspiring Excellence

**CSE370 : Database Systems**  
**Project Report**  
**Project Title : SAFR**

| Group No : _06_, CSE370 Lab Section : _10_, Spring 2026 |                            |                                                                                          |
|---------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------|
| ID                                                      | Name                       | Contribution                                                                             |
| 24101513                                                | Tahmeed Labib Khan Majhlis | EER diagram(refugees, NGOS,), schema primary and foreign connection, db code, 4 features |
| 24101211                                                | Wahid Al Amin              | EER diagram (Volunteer),EER participation and cardiarity , schema half, 3 feature        |
| 23101470                                                | Tasneem Akram Nawshin      | EER diagram (medical, donor),EER participation and cardiarity, schema half, 2 features   |

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## Introduction

[Briefly describe the idea of your project]

**Aim: To help those in need find a safer place and find their loved ones**

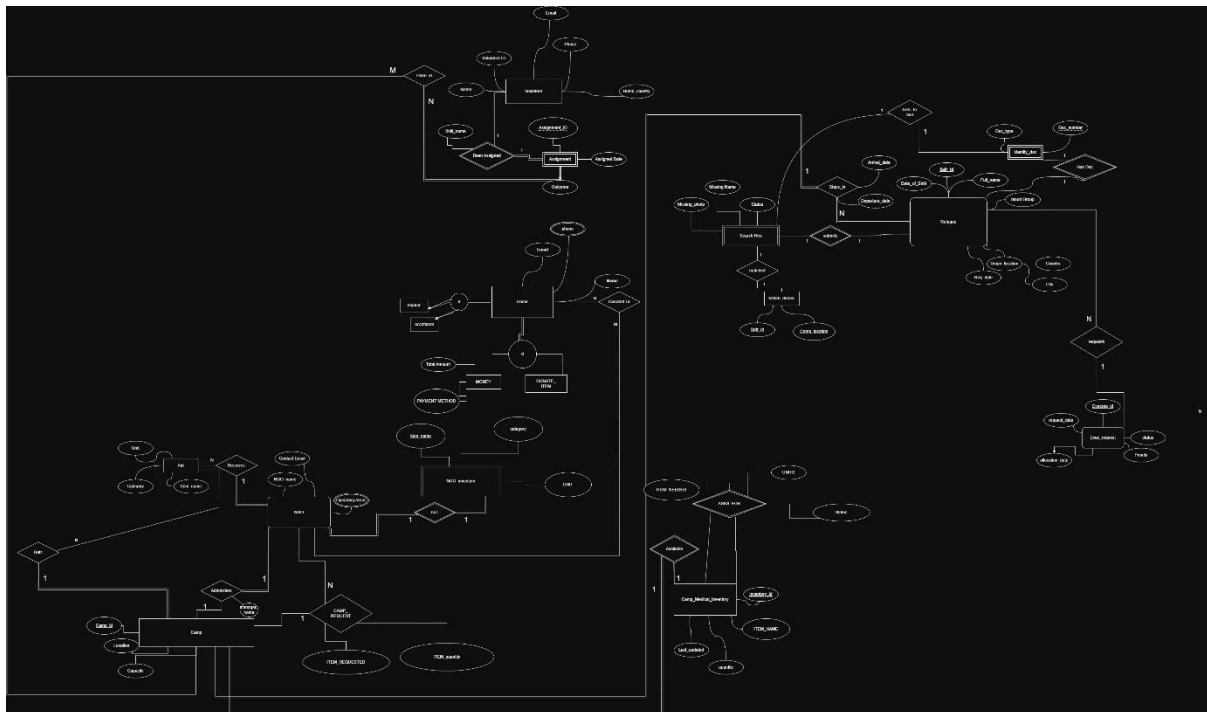
As war rages on, families need to leave their homes to keep themselves safe. But more than often they fail to find shelter or find where they have everything except what they actually need. So they keep moving but in search of what they actually need. This project can resolve this problem.

## Project Features

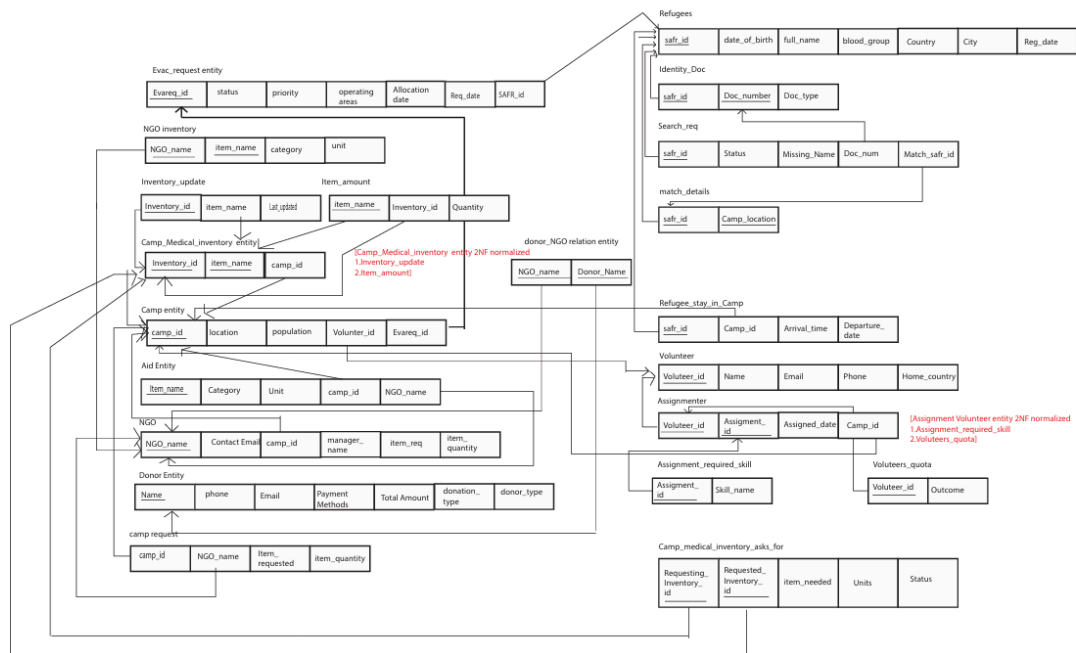
| ID,<br>Name | Features [3 per member] |
|-------------|-------------------------|
|-------------|-------------------------|

|                                               |      |                                                                      |
|-----------------------------------------------|------|----------------------------------------------------------------------|
| 24101513,<br>Tahmeed<br>Labib Khan<br>Majhlis | Ft 1 | Refugee & Displacement Tracking System                               |
|                                               | Ft 2 | Civilian Evacuation Transportation                                   |
|                                               | Ft 3 | Finding family members through fingerprints or Identification papers |
|                                               | Ft 4 | Camp change                                                          |
| 24101211,<br>Wahid Al<br>Amin                 | Ft 1 | Accepting camp request                                               |
|                                               | Ft 2 | Adding aid inventory                                                 |
|                                               | Ft 3 | Crisis Volunteer & Skill Matching Platform                           |
| 23101470,<br>Tashneem<br>Nawshin              | Ft 1 | Medical Resource Management                                          |
|                                               | Ft 2 | Donations                                                            |
|                                               | Ft 3 |                                                                      |

## ER/EER Diagram



# Schema Diagram



## Normalization

- Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
- Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
- Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

- No 1NF
- 2 2NF, there is no partial functional dependencies in our relational schema
- No 3NF

## Frontend Development

Briefly discuss about frontend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

**Contribution of ID : 24101211, Name : Wahid Al Amin:** Designed the frontend of volunteer and NGO in volunteer there are assignment values and outcomes input button. In NGO there are link buttons to go to add assignment, aidview with adding aid, requesting camp request button where there will be details to ask for items that are needed by the current user's camp. I have made dynamic NGO designs to be able to accept those camp requests. I have also included a feature to update the Aid Inventory of the NGO. I have maintained a tabular structure to ease the eyes of the user. Finally, I have created the volunteer pages to find volunteers who have the needed skills to work for the benefit of the NGO/Camps. I have kept a form system and a dropdown system here as well to ensure the easy access of the users. I have also made it possible to report outcomes of the volunteers who have finished working on their assignments. The entire frontend of the modules that I worked in was designed keeping easy usability of the users in mind which is why I have kept it simple and used colours that are comfortable to the eyes.

**Contribution of ID : 24101513, Name : Tahmeed Labib Khan Majhlis:** The frontend of the refugee module is designed like a simple dashboard-style web app where everything revolves around the logged-in user. There's a header with the SAFR logo and a logout option followed by a nav bar that allows refugees to within Dashboard, Evac Requests, Camp Change, Camp History etc. The dashboard shows personal details and current camp information while the other features allow uploading document ids. Forms were used where interaction is needed for instance - submitting evacuation requests, searching for family members or updating camp locations. I've also kept dropdowns for easy access. Overall, the frontend focuses on clarity and usability which makes the refugees easily view their information, submit requests and track updates without confusion.

**Contribution of ID : 23101470, Name : Tasneem Akram Nawshin:**

**NGO Medical Request:** This page features a top bar displaying the SAFR logo and name of the NGO account that was logged in with a logout button, followed by a navigation menu where "Medical Aid" is highlighted as the active section. The main content area presents a form for creating medical supply requests, including a read-only field for the camp ID, a dropdown menu to select an inventory to request from (showing the inventory ID, item name, and camp ID), and a number input for required units. A submit button triggers the request. Below the form, two tables display the NGO's outgoing medical requests: one for pending requests with a "Pending" status badge, and another for accepted requests with an "Accepted" badge. Each table lists the requesting inventory ID, requested inventory ID, item needed,

units, the camp being requested from, and the status. Success or error messages appear above the form upon submission, and the entire page uses custom SAFR styling to ensure visual consistency. I only used simple html and custom css to make this feature.

**Donor:** The donor portion features a frontend built with simple html, custom and responsive css. At the top, a dark navy blue topbar displays the "SAFR — Donor Portal" branding on the left and, if logged in, shows the donor's name with a logout link on the right. Below that, a blue navigation menu dynamically adjusts based on login state: for logged-out users, it shows "Login" and "Sign Up" links; for logged-in users, it displays "Donate," "My Profile," and "Logout." All the forms, whether text fields, email, password, number inputs, or dropdown menus, use proper padding and borders, with a nice blue highlight when you click into them. When something goes right, you will see a green message tab. If there is an error, a red tab pops up instead. Once you are logged in and head to the "Donate" page, you will find a dropdown to pick an NGO, a box to enter your donation amount, and a menu to choose your payment method, plus a short sentence explaining where your money will go. The "My Profile" page shows your saved information in a light blue box and lists all the NGOs you have supported so far. Buttons throughout the site are in solid blue.

## Backend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

### Contribution of 24101211, Name : Wahid Al Amin

**Volunteer:** the backend handles all volunteers list from the database. In addition, in the outcome, the quota will be sent to the volunteer quota.

**NGO:** the backend controls the show of all NGOS in the database. In addition, in the assignment tab, it will show all assignments assigned and will add assignments to volunteers based on skills. In aidview , it will show all aids from the database. moreover , there can be items added and accept request then subtract from NGO inventory

### Contribution of ID : 24101513, Name : Tahmeed Labib Khan Majhlis

Worked on all the refugee related features. My works are in refugee registration, identification, evacuation and family search. I also worked on the ngo side on one feature where the designated NGO can see the evac request and can decide to accept it or not according to the priority of the list.

A refugee can also use the live search button to find if a family member is in a refugee camp or not. They cant request more than one evacuation at a time.

### Contribution of ID :23101470, Name : Tasneem Akram Nawshin:

**Donor:** The backend of the donor portal handles everything related to donor registration, login, and donation processing. When a donor signs up, the system saves their personal information like name, email, and phone number into the donor table, while storing their login credentials separately in the donor credential table for security. During login, the system checks whether the entered name and password match what is stored before allowing access. When a donor makes a donation, the backend first validates the donation details such as ensuring the name is not empty, the email format is correct, and the amount is a positive number. If everything looks good, it updates the donor's total donated amount, records their chosen payment method, and creates a link between the donor and the chosen NGO in a separate table to keep track of which NGOs each donor has supported.

**Medical:** For the medical aid page, the backend focuses on managing requests for medical supplies between different camps. When an NGO wants to request medical items, the system first identifies which camp they belong to by looking up their NGO name. It then retrieves

the inventory ID associated with that camp. The backend checks that the requested inventory exists and has a valid item name before inserting a new request into the database. This request record includes the requested inventory ID, the requested inventory ID, the name of the item needed, the number of units required, and a status flag where zero means pending and one means accepted. The system also fetches all pending and accepted requests for display, joining multiple tables to show which camp is requesting from which other camp. This allows NGOs to track their outgoing requests and see whether other camps have accepted their requests for medical supplies.

## Source Code Repository

Upload source code to GitHub or Google Drive and share a publicly accessible link in this section of the report.

<https://github.com/Horuseus/CSE-370-PROJECT>

## Conclusion

As a result, we can see that through our website, the refugees can meet their missing loved people. In addition, they can do evacuation requests at any time. Moreover, the NGOs can access the data files about available aids and provide them to the requested camp. Furthermore, the organization can get the volunteers based on the skills without calling them physically.

## References

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